Two Ceiling Pisot Sequences Satisfying a Fifth-Order Recurrence

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Abstract

We prove the conjectured fifth-order linear recurrences for the ceiling Pisot sequences starting with 8, 9 and with 9, 10. The two sequences differ by one at every index. After subtracting a constant, both reduce to the same fourth-order recurrence. Its companion matrix induces a six-dimensional recurrence on determinants of consecutive state vectors. An exact bound for the fourteenth power of this matrix yields an induction that proves the ceiling inequalities at every index. The proof identifies the unique equality case for the first sequence. We also determine which integer translates satisfy the ceiling rule and derive the limiting rounding errors from the zeros of the fourth-order characteristic polynomial.

1 Introduction

For positive integers $x < y$, let $L(x, y) = (a_n)_{n\geq 0}$ be the *ceiling Pisot sequence* defined by

$$a_0 = x, \qquad a_1 = y, \qquad a_n = \left\lceil \frac{a_{n-1}^2}{a_{n-2}} \right\rceil \quad (n \geq 2). \tag{1}$$

The sequences $L(8, 9)$ and $L(9, 10)$ are A048590 and A048592, respectively, in the On-Line Encyclopedia of Integer Sequences [4, 5]. Their entries record the conjecture that both satisfy

$$a_n = 2a_{n-1} - a_{n-2} + a_{n-4} - a_{n-5} \quad (n \geq 6). \tag{2}$$

Theorem 1. *For $t \in \{3,4\}$, let $(a_n^{(t)})_{n\geq 0} = L(t+5, t+6)$. Both sequences satisfy (2) for every $n \geq 6$, and*

$$a_n^{(4)} = a_n^{(3)} + 1 \quad (n \geq 0).$$

Furthermore, for $t \in \{3,4\}$,

$$0 \leq a_n^{(t)} a_{n-2}^{(t)} - (a_{n-1}^{(t)})^2 < a_{n-2}^{(t)} \quad (n \geq 2).$$

Equality in the lower bound occurs exactly when $t = 3$ and $n = 6$.

The proof is given in Section 4, after the necessary lemmas.

A linear recurrence suggested by a long initial segment need not persist. For the nearest-integer convention, Boyd [1] proved that some Pisot sequences satisfy no linear recurrence. He later studied which characteristic polynomials occur for recurrent Pisot sequences [2]. These results place the recognition of a particular recurrence within a broader problem: the rounding rule may produce either linear behavior or a sequence for which no such relation exists.

Ekhad, Sloane, and Zeilberger [3] developed automated methods for proving or disproving proposed recurrences. Their argument expresses consecutive determinants in terms of pairwise products of characteristic zeros, bounds their growth relative to the candidate sequence, and checks the remaining finite range. They also gave examples in which an apparent recurrence fails after thousands of terms. The present proof uses the same determinant comparison, expressed through a matrix acting on all two-by-two minors of consecutive state vectors.

The rounding convention requires separate attention. For the ceiling rule, the necessary and sufficient condition is

$$0 \leq a_n a_{n-2} - a_{n-1}^2 < a_{n-2}, \tag{3}$$

provided $a_{n-2} > 0$. The lower endpoint is included. In contrast, nearest-integer rounding permits determinants of either sign. Replacing the ceiling by $\lfloor s + 1 \rfloor$ would also change the rule when s is an integer. This distinction matters here because the determinant for $L(8,9)$ vanishes at index 6.

The characteristic polynomial in (2) factors as

$$x^5 - 2x^4 + x^3 - x + 1 = (x-1)(x^4 - x^3 - 1).$$

Subtracting a constant from either candidate leaves the same fourth-order sequence. Its consecutive determinants form part of a six-dimensional linear

system. A bound for the fourteenth power of its matrix, compared with a lower growth bound for the fourth-order sequence, gives an induction over fourteen residue classes. The resulting certificate proves (3) using integer arithmetic and rational inequalities. The initial cases account for the single equality at index 6.

We also determine all integer translates of the common sequence that satisfy the ceiling rule, obtain the two generating functions, and compute the limiting rounding errors. The zeros of $x^4 - x^3 - 1$ are needed only for this last analysis; the proof of the ceiling identities is elementary.

2 The common fourth-order sequence

Define $(b_n)_{n\ge1}$ by

$$(b_1, b_2, b_3, b_4) = (6, 8, 11, 15), \qquad b_n = b_{n-1} + b_{n-4} \quad (n \ge 5). \tag{4}$$

For an integer t, put

$$u_0^{(t)} = t + 5, \qquad u_n^{(t)} = b_n + t \quad (n \ge 1). \tag{5}$$

No value of b_0 is used in this definition.

For $t \in \{3, 4\}$, the sequence $u^{(t)}$ is a candidate for $L(t + 5, t + 6)$. The proof of Theorem 1 establishes this identification.

For $t = 3$, the first six terms are $8, 9, 11, 14, 18, 24$. Adding one gives the initial terms for $t = 4$. For every integer t, (4) gives

$$u_n^{(t)} = u_{n-1}^{(t)} + u_{n-4}^{(t)} - t \quad (n \ge 5). \tag{6}$$

Subtracting consecutive instances of (6) proves (2) for these candidates.

Conversely, suppose $(c_n)_{n\ge0}$ has initial terms $8, 9, 11, 14, 18, 24$ and satisfies (2) for $n \ge 6$. Set $f_n = c_n - c_{n-1} - c_{n-4}$ for $n \ge 5$. The fifth-order recurrence gives $f_n = f_{n-1}$ for $n \ge 6$, and $f_5 = -3$. Thus $c_n - 3$ satisfies (4) for $n \ge 1$, so $c_n = u_n^{(3)}$ for every $n \ge 0$. This reduction does not extend to $n = 0$: continuing (4) backwards would give $b_0 = 4$, whereas $c_0 - 3 = 5$.

Lemma 2. *The sequence $(b_n)_{n\ge1}$ is positive and strictly increasing, and*

$$b_j \le 2b_{j-1} \quad (j \ge 2).$$

For every $m \ge 4$,

$$b_{m+14} \ge \frac{141}{2} b_m, \qquad b_{m+6} \ge \frac{43}{8} b_m. \tag{7}$$

Proof. The four initial values are positive and increasing. For $n \geq 5$, (4) gives $b_n - b_{n-1} = b_{n-4} > 0$ inductively. The inequality $b_j \leq 2b_{j-1}$ holds for $2 \leq j \leq 4$ by direct substitution. For $j \geq 5$, it follows from $b_j = b_{j-1} + b_{j-4} \leq 2b_{j-1}$. Consequently,

$$b_{m-j} \geq 2^{-j} b_m \quad (m \geq 4,\ 0 \leq j \leq 3). \tag{8}$$

For $n \geq 4$, define

$$x_n = (b_n, b_{n-1}, b_{n-2}, b_{n-3})^{\mathsf{T}}, \qquad A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Then $x_{n+1} = Ax_n$. Integer multiplication gives

$$A^6 = \begin{pmatrix} 4 & 1 & 2 & 3 \\ 3 & 1 & 1 & 2 \\ 2 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \tag{9}$$

$$A^{14} = \begin{pmatrix} 50 & 19 & 26 & 36 \\ 36 & 14 & 19 & 26 \\ 26 & 10 & 14 & 19 \\ 19 & 7 & 10 & 14 \end{pmatrix}. \tag{10}$$

The first rows of (9) and (10), together with (8), give

$$b_{m+6} \geq \left(4 + \frac{1}{2} + \frac{2}{4} + \frac{3}{8}\right) b_m = \frac{43}{8} b_m,$$
$$b_{m+14} \geq \left(50 + \frac{19}{2} + \frac{26}{4} + \frac{36}{8}\right) b_m = \frac{141}{2} b_m.$$

□

3 The determinant induction

Index the coordinates of x_n by $0, 1, 2, 3$. For column vectors $v, w \in \mathbb{R}^4$, write $v \wedge w \in \mathbb{R}^6$ for the vector with coordinates $v_i w_j - v_j w_i$, ordered as

$$(01, 02, 03, 12, 13, 23).$$

Set $W_n = x_n \wedge x_{n-1}$ for $n \geq 5$, and let

$$H_n = b_n b_{n-2} - b_{n-1}^2.$$

The first coordinate of W_n is H_n. The initial vectors are

$$\begin{aligned} x_4 &= (15, 11, 8, 6)^{\mathsf{T}}, \qquad x_5 = (21, 15, 11, 8)^{\mathsf{T}}, \\ W_5 &= (6, 3, 6, -1, 2, 2)^{\mathsf{T}}. \end{aligned}$$

For any 4 by 4 matrix B, expansion of a minor gives

$$\begin{aligned} &(Bv)_i(Bw)_j - (Bv)_j(Bw)_i \\ &\qquad = \sum_{0 \leq k < l \leq 3} (B_{ik}B_{jl} - B_{il}B_{jk})(v_k w_l - v_l w_k). \end{aligned}$$

Applying this identity to $B = A$ yields

$$W_{n+1} = MW_n \quad (n \geq 5), \qquad M = \begin{pmatrix} 0 & 0 & -1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}. \tag{11}$$

Thus M is the second exterior power of A in the stated coordinate order. Equivalently, it maps a vector ξ to

$$(-\xi_{03},\ \xi_{01} - \xi_{13},\ \xi_{02} - \xi_{23},\ \xi_{01},\ \xi_{02},\ \xi_{12})^{\mathsf{T}}.$$

This formula also specifies the signs without exterior-power notation.

Let $\|\xi\|_\infty$ denote the largest absolute coordinate of a vector $\xi \in \mathbb{R}^6$.

Lemma 3. *For every $n \geq 20$,*

$$\|W_n\|_\infty \leq b_{n-8}. \tag{12}$$

Proof. Repeated multiplication of the matrix in (11) gives

$$M^{14} = \begin{pmatrix} 16 & 14 & 4 & -3 & -10 & -8 \\ 6 & 24 & 14 & 6 & 1 & -10 \\ -11 & 6 & 16 & 8 & 14 & 4 \\ -4 & 10 & 8 & 6 & 6 & -3 \\ -14 & -1 & 10 & 7 & 14 & 6 \\ -8 & -6 & 3 & 2 & 7 & 6 \end{pmatrix}. \tag{13}$$

n	$\|W_n\|_\infty$	b_{n-8}
20	192	200
21	244	276
22	373	381
23	373	526
24	688	726
25	757	1002
26	1055	1383
27	1294	1909
28	1364	2635
29	2121	3637
30	2121	5020
31	4236	6929
32	4690	9564
33	6440	13201

Table 1: The fourteen initial cases for the determinant bound.

The absolute row sums in (13) are $55, 61, 59, 37, 52, 32$. Therefore

$$\|M^{14}\xi\|_\infty \le 61\|\xi\|_\infty \quad (\xi \in \mathbb{R}^6).$$

Iterating (11) now gives

$$\|W_{n+14}\|_\infty \le 61\|W_n\|_\infty \quad (n \ge 5). \tag{14}$$

The block length fourteen is useful because 61 is smaller than the lower growth factor $141/2$ in Lemma 2.

The fourteen initial inequalities in Table 1 follow from (4) and $W_n = M^{n-5}W_5$. Suppose (12) holds at an index $n \ge 20$. Since $n - 8 \ge 12$, Lemma 2 and (14) give

$$\|W_{n+14}\|_\infty \le 61 b_{n-8} < \frac{141}{2} b_{n-8} \le b_{n+6} = b_{(n+14)-8}.$$

Every integer $n \ge 20$ has a unique representation $n = 20 + s + 14k$ with $0 \le s \le 13$ and $k \ge 0$. Induction on k, starting with Table 1, proves the lemma. □

4 Proof of the ceiling identities

Theorem 1 (restated). *For $t \in \{3,4\}$, let $(a_n^{(t)})_{n\geq 0} = L(t+5, t+6)$. Both sequences satisfy (2) for every $n \geq 6$, and*

$$a_n^{(4)} = a_n^{(3)} + 1 \quad (n \geq 0).$$

Furthermore, for $t \in \{3,4\}$,

$$0 \leq a_n^{(t)} a_{n-2}^{(t)} - (a_{n-1}^{(t)})^2 < a_{n-2}^{(t)} \quad (n \geq 2).$$

Equality in the lower bound occurs exactly when $t = 3$ and $n = 6$.

Proof of Theorem 1. For $t \in \{3,4\}$ and $n \geq 2$, define

$$D_t(n) = u_n^{(t)} u_{n-2}^{(t)} - (u_{n-1}^{(t)})^2.$$

For $n \geq 9$, two applications of (4) give

$$\begin{aligned} b_n + b_{n-2} - 2b_{n-1} &= (b_n - b_{n-1}) - (b_{n-1} - b_{n-2}) \\ &= b_{n-4} - b_{n-5} = b_{n-8}. \end{aligned}$$

Expanding the determinant consequently gives the exact identity

$$D_t(n) = H_n + t b_{n-8} \quad (n \geq 9). \tag{15}$$

Lemma 2 shows that every term of $u^{(3)}$ and $u^{(4)}$ is positive. For $n \geq 20$, Lemma 3 gives $|H_n| \leq b_{n-8}$. By (15),

$$0 < (t-1) b_{n-8} \leq D_t(n) \leq (t+1) b_{n-8}. \tag{16}$$

The other growth bound in (7), applied at $m = n - 8$, implies

$$D_t(n) \leq 5 b_{n-8} < \frac{43}{8} b_{n-8} \leq b_{n-2} < u_{n-2}^{(t)}.$$

This proves both required inequalities for every $n \geq 20$.

For $2 \leq n \leq 19$, Table 2 gives the determinant and its upper bound by direct evaluation of (4) and (5). Every row satisfies $0 \leq D_t(n) < u_{n-2}^{(t)}$. The only zero is

$$D_3(6) = 32 \cdot 18 - 24^2 = 0.$$

n	$D_3(n)$	$u^{(3)}_{n-2}$	$D_4(n)$	$u^{(4)}_{n-2}$
2	7	8	8	9
3	5	9	6	10
4	2	11	3	12
5	12	14	14	15
6	0	18	2	19
7	8	24	11	25
8	7	32	11	33
9	33	43	39	44
10	23	58	31	59
11	28	79	39	80
12	20	108	35	109
13	83	148	104	149
14	111	203	140	204
15	135	279	175	280
16	95	384	150	385
17	204	529	280	530
18	369	729	474	730
19	564	1005	709	1006

Table 2: The ceiling inequalities before the induction threshold.

The strict lower bound in (16) excludes any further zero.

For integers h, p and a positive integer q,

$$h = \left\lceil \frac{p}{q} \right\rceil \quad \Longleftrightarrow \quad h - 1 < \frac{p}{q} \leq h \quad \Longleftrightarrow \quad 0 \leq hq - p < q.$$

Apply this equivalence with $h = u^{(t)}_n$, $p = (u^{(t)}_{n-1})^2$, and $q = u^{(t)}_{n-2}$. The verified inequalities show that $u^{(t)}$ satisfies (1) at every $n \geq 2$. Its initial terms are $t + 5, t + 6$. Sequential evaluation of (1) determines each subsequent term uniquely, so $u^{(t)} = L(t + 5, t + 6)$. The fifth-order recurrence was established in Section 2. Finally, (5) gives $u^{(4)}_n = u^{(3)}_n + 1$ for every $n \geq 0$, proving the asserted difference. □

5 Integer translates and generating functions

Corollary 4. *Let* $(a_n)_{n\geq 0} = L(8,9)$ *and* $(\widetilde{a}_n)_{n\geq 0} = L(9,10)$. *Then* $\widetilde{a}_n = a_n + 1$ *for every* $n \geq 0$.

Proof. By the proof of Theorem 1, these sequences are $u^{(3)}$ and $u^{(4)}$. Their difference is one at every index by (5). □

Proposition 5. *Let* $t \geq -4$ *be an integer, and define* $u^{(t)}$ *by* (5). *Then* $u^{(t)} = L(t+5, t+6)$ *if and only if* $t \in \{3,4\}$.

Proof. The initial pair $t+5, t+6$ is positive and increasing precisely when $t \geq -4$. For these values of t, all subsequent terms are positive since $b_n \geq 6$ by Lemma 2. If the ceiling rule holds, its determinant at $n = 6$ must be nonnegative. Since $(b_4, b_5, b_6) = (15, 21, 29)$,

$$(29+t)(15+t) - (21+t)^2 = 2t - 6 \geq 0,$$

so $t \geq 3$. At $n = 5$, the strict upper bound gives

$$(21+t)(11+t) - (15+t)^2 = 2t + 6 < 11 + t,$$

so $t < 5$. The converse follows from the proof of Theorem 1. □

To index the generating function from zero, put $d_k = b_{k+1}$ for $k \geq 0$. Multiplying its formal power series by $1 - z - z^4$ and using (4) gives

$$F(z) := \sum_{k\geq 0} d_k z^k = \frac{6 + 2z + 3z^2 + 4z^3}{1 - z - z^4}. \tag{17}$$

It follows that

$$\sum_{n\geq 0} u_n^{(t)} z^n = 5 + \frac{t}{1-z} + \frac{z(6 + 2z + 3z^2 + 4z^3)}{1 - z - z^4}.$$

This identity holds formally for every integer t. For $t = 3, 4$ it is the generating function of the corresponding ceiling Pisot sequence.

6 Limiting rounding errors

Let $Q(x) = x^4 - x^3 - 1$. We first give rational bounds for its zeros.

Lemma 6. *The polynomial Q has four simple zeros. Its two real zeros $\lambda > 0$ and $\mu < 0$ satisfy*

$$\frac{69}{50} < \lambda < \frac{7}{5}, \qquad -\frac{41}{50} < \mu < -\frac{81}{100}.$$

Every zero $r \neq \lambda$ satisfies

$$\frac{4}{5} < |r| < \frac{19}{20}, \qquad |4r - 3| > 2.$$

Proof. The derivative is $Q'(x) = x^2(4x-3)$. Thus Q is strictly decreasing up to $3/4$ and strictly increasing thereafter. Since $Q(3/4) = -283/256$, it has exactly two real zeros. Neither 0 nor $3/4$ is a zero of Q, so all four zeros are simple. Exact substitution gives

$$\begin{aligned} Q(69/50) &= -8329/6250000, & Q(7/5) &= 61/625, \\ Q(-41/50) &= 21811/6250000, & Q(-81/100) &= -3809179/100000000. \end{aligned}$$

These signs give the stated real intervals.

Write the nonreal zeros as $r = \alpha + i\beta$ and $\overline{r}$. Vieta's formulas imply

$$\alpha = \frac{1 - \lambda - \mu}{2} < \frac{11}{50}, \qquad \frac{250}{287} < |r|^2 = -\frac{1}{\lambda\mu} < \frac{5000}{5589}.$$

The last interval is contained in $((4/5)^2, (19/20)^2)$. Also,

$$|4r - 3| \geq |4\alpha - 3| = 3 - 4\alpha > \frac{53}{25} > 2.$$

The same modulus and denominator bounds for μ follow directly from its real interval. □

Lemma 7. *For every $k \geq 0$,*

$$d_k = \sum_{Q(r)=0} C(r) r^k, \qquad C(r) = \frac{6r^3 + 2r^2 + 3r + 4}{r^2(4r - 3)}. \tag{18}$$

In particular, if $C = C(\lambda)$ and $\rho = 19/20$, then

$$d_k = C\lambda^k + e_k, \qquad C > 2, \qquad |e_k| < 36\rho^k. \tag{19}$$

Proof. Let $N(z) = 6 + 2z + 3z^2 + 4z^3$ and $G(z) = 1 - z - z^4$. The simple poles of (17) are $1/r$, where $Q(r) = 0$. The coefficient of $(1 - rz)^{-1}$ in the partial fraction expansion is

$$-\frac{rN(1/r)}{G'(1/r)} = \frac{rN(1/r)}{4r - 3} = C(r).$$

Here $G'(1/r) = -(4r-3)$ follows from $r^4 = r^3+1$. Expanding each $(1-rz)^{-1}$ as a formal power series proves (18).

The positive zero satisfies $1 < \lambda < 3$, so

$$C(\lambda) > \frac{6\lambda}{4\lambda - 3} > 2.$$

For each of the other three zeros, Lemma 6 gives

$$|C(r)| < \frac{6 + 2 + 3 + 4}{(4/5)^2 \cdot 2} = \frac{375}{32} < 12.$$

Their combined contribution therefore has absolute value less than $36\rho^k$, proving (19). □

Proposition 8. *For $t \in \{3, 4\}$,*

$$\lim_{n\to\infty} \left(u_n^{(t)} - \frac{(u_{n-1}^{(t)})^2}{u_{n-2}^{(t)}} \right) = t(\lambda - 1)^2.$$

Proof. For $n \geq 3$, put $k = n - 3$, $T_k = C\lambda^k$, and $\eta_j = e_{k+j}$ for $j = 0, 1, 2$. Then

$$\begin{aligned} u_{n-2}^{(t)} &= T_k + t + \eta_0, \\ u_{n-1}^{(t)} &= \lambda T_k + t + \eta_1, \\ u_n^{(t)} &= \lambda^2 T_k + t + \eta_2. \end{aligned}$$

Expanding the determinant gives

$$\begin{aligned} \frac{D_t(n)}{T_k} &= t(\lambda - 1)^2 + \eta_2 + \lambda^2\eta_0 - 2\lambda\eta_1 \\ &\qquad + \frac{t(\eta_2 + \eta_0 - 2\eta_1) + \eta_0\eta_2 - \eta_1^2}{T_k}. \end{aligned}$$

By (19), each η_j tends to zero and T_k tends to infinity. Thus $D_t(n)/T_k$ tends to $t(\lambda - 1)^2$, while $u_{n-2}^{(t)}/T_k$ tends to one. Dividing these limits proves the assertion. □

Exact rational substitution into Q gives the finer enclosure

$$\frac{1380277569097}{10^{12}} < \lambda < \frac{1380277569098}{10^{12}}.$$

Since $t(x-1)^2$ is increasing for $x > 1$ and $t = 3, 4$, this enclosure gives the limiting errors

$$3(\lambda - 1)^2 \approx 0.4338330887, \qquad 4(\lambda - 1)^2 \approx 0.5784441182,$$

for A048590 and A048592, respectively. Both values are rounded to ten decimal places.

7 Computational verification

The finite calculations used in the determinant proof are the powers A^6, A^{14}, and M^{14}, the absolute row sums of M^{14}, the two growth constants, and the entries of Tables 1 and 2. The table ranges are $20 \le n \le 33$ and $2 \le n \le 19$, respectively. Each entry follows by integer matrix multiplication or iteration of (4); no numerical root approximation is used in these calculations.

The supplementary programs implement these checks using Python integers and rational numbers. Separate implementations reconstruct M from the minors of A and compute the base vectors directly from the sequence. They also verify the rational evaluations in Section 6. A source checker compares the displayed matrices and tables with the computed values. The accompanying certificates and instructions specify the inputs and commands needed to reproduce the calculations.

As an additional check, direct iteration of the nonlinear rule agrees with each linear candidate through index 1000. This comparison is supplementary: the induction in Lemma 3, together with the finite ceiling cases, proves the identities for all indices.

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